

The effect of marjoram (Origanum majorana) tea on the hormonal profile of women with polycystic ovary syndrome: a randomised controlled pilot study.



BACKGROUND: In traditional medicine, marjoram herb (*Origanum majorana*) is locally reputed for its ability to restore hormonal balance and to regulate the menstrual cycle. Therefore, this pilot study aimed to investigate the effects of marjoram tea on the hormonal profile of women with polycystic ovary syndrome (PCOS) in a randomised, double-blind, placebo-controlled trial. **METHODS:** Twenty-five patients were assigned to receive marjoram tea or a placebo tea twice daily for 1 month (intervention group: n = 14; placebo group: n = 11). The hormonal and metabolic parameters measured at baseline, as well as after the intervention, were: follicle-stimulating hormone, luteinising hormone, progesterone, oestradiol, total testosterone, dehydroepiandrosterone-sulphate (DHEA-S), fasting insulin and glucose, homeostasis model assessment for insulin resistance (HOMA-IR) and glucose to insulin ratio. **RESULTS:** Marjoram tea significantly reduced DHEA-S and fasting insulin levels ($P < 0.05$) by a mean (SD) of 1.4 (0.5) $\mu\text{mol L}^{-1}$ and 1.9 (0.8) $\mu\text{U mL}^{-1}$, respectively. In comparison to the placebo group, the change was only significant for DHEA-S ($P = 0.05$) but not for insulin ($P = 0.08$). HOMA-IR was not reduced significantly in the intervention group ($P = 0.06$), although the change was significant compared to the placebo group ($P < 0.05$). **CONCLUSIONS:** The results obtained in the present study show the beneficial effects of marjoram tea on the hormonal profile of PCOS women because it was found to improve insulin sensitivity and reduce the levels of adrenal androgens. Further research is needed to confirm these results and to investigate the active components and mechanisms contributing to such potential beneficial effects of marjoram herb. © 2015 The British Dietetic Association Ltd.